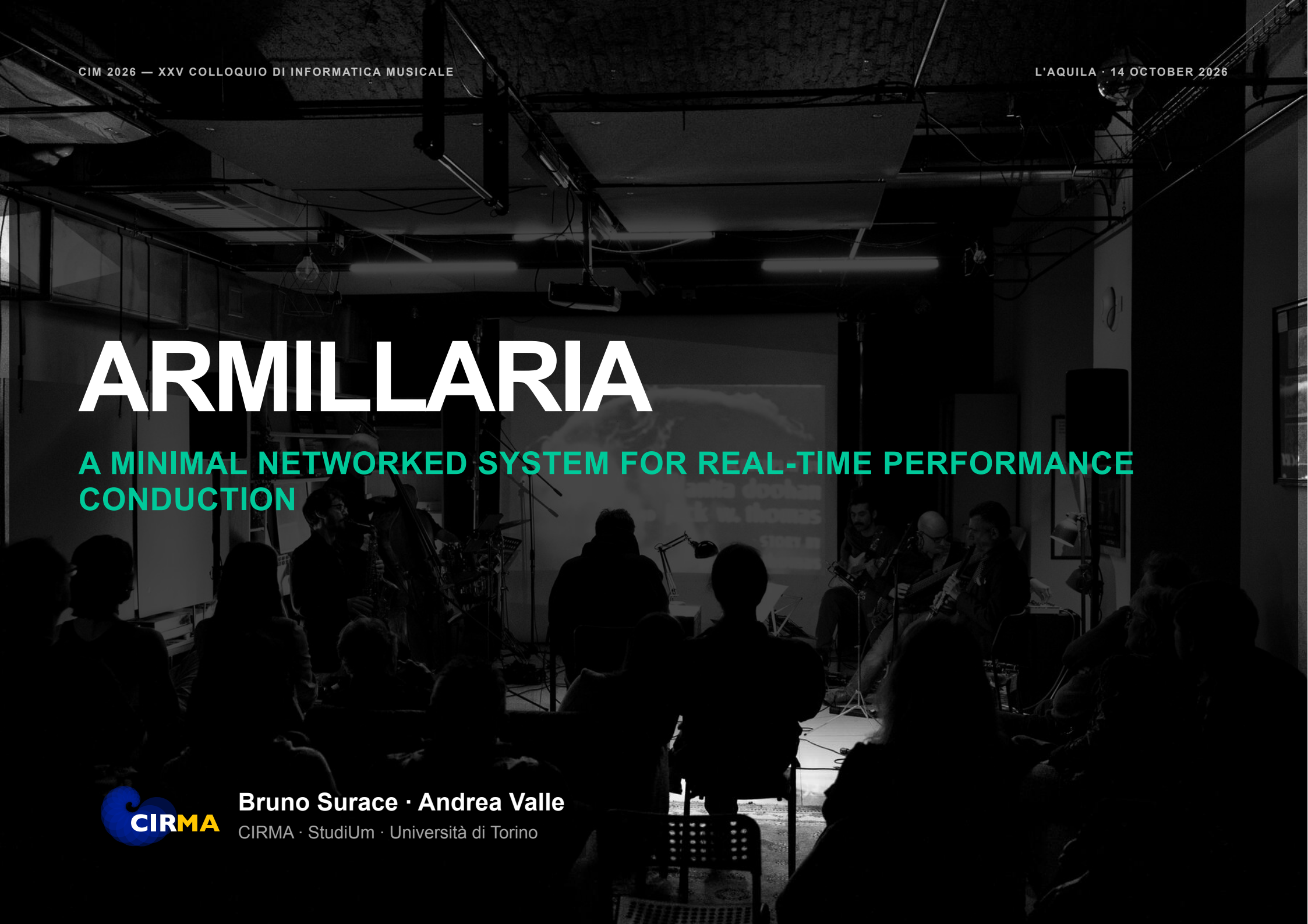


ARMILLARIA

A MINIMAL NETWORKED SYSTEM FOR REAL-TIME PERFORMANCE CONDUCTION

Bruno Surace · Andrea Valle

CIRMA · StudiUm · Università di Torino

01

NOTATION & CONDUCTION

THE IDEA

A conductor — the *kybernete* — types plain-text instructions that appear instantly on musicians' browsers over a local WiFi network. No apps, no accounts, no proprietary hardware.

The name borrows from Armillaria, the honey fungus genus whose mycelium spreads in vast underground networks — invisible at the surface, connecting many nodes without a centre.

FOUR AXES OF NOTATION

— **Type** — standard, graphic, or *textual*

— **Orientation** — representational or *prescriptive*

— **Timing** — outside-time or *in-time*

— **Addressee** — machine, or *human*

SYSTEM	TYPE	ORIENT.	TIMING	ADDR.
Traditional score	std/graph	repr.	out	hum.
Decibel SP	graphic	mixed	in	hum.
FlowScore	standard	mixed	in	hum.
MIDI (file)	standard	presc.	out	mach.
Live coding	textual	presc.	in	mach.
Armillaria	**textual**	**presc.**	**in**	**hum.**

LIVE CODING, DISPLACED

> *A live coder who types freq = 440 produces a deterministic result; a kybernete who types pianissimo produces as many results as there are musicians reading it.*

Armillaria — textual, prescriptive, in-time, *human-addressed*

REFERENCES

NOTATION

1 A. Valle, *Contemporary Music Notation*, 2024

2 A. Valle, "Notazioni elettromeccaniche, o forse no," 2019

3 C. Cardew, *Treatise Handbook*, 1971

4 C. Hope et al., "The Decibel ScorePlayer," TENOR 2015

5 C. Hope, "The future is graphic," *Organised Sound*, 2020

6 J. Cage & A. Knowles, *Notations*, 1969

7 T. Sauer, *Notation 21*, 2009

9 T. T. Stanley, *Butch Morris and the Art of Conduction*, 2009

10 L. D. Morris, *The Art of Conduction*, 2017

LIVE CODING

11 N. Collins et al., "Live Coding in Laptop Performance," 2003

12 T. Magnusson, "Algorithms as Scores," 2011

13 A. Valle, "Live Writing with the Rumentarium," 2011

NETWORKED PERFORMANCE

16 S. Gresham-Lancaster, "The Aesthetics and History of the Hub," 1998

17 A. Barbosa, "Displaced Soundscapes," 2003

18 G. Weinberg, "Interconnected Musical Networks," 2005

19 L. Turchet et al., "Internet of Musical Things," 2018

20 J. Freeman & A. Van Troyer, "Collaborative Textual Improvisation," 2011

21 G. Hajdu, "Quintet.net," 2005

22 P. Louzeiro, "Comprovisador," TENOR 2018

02

SYSTEM ARCHITECTURE

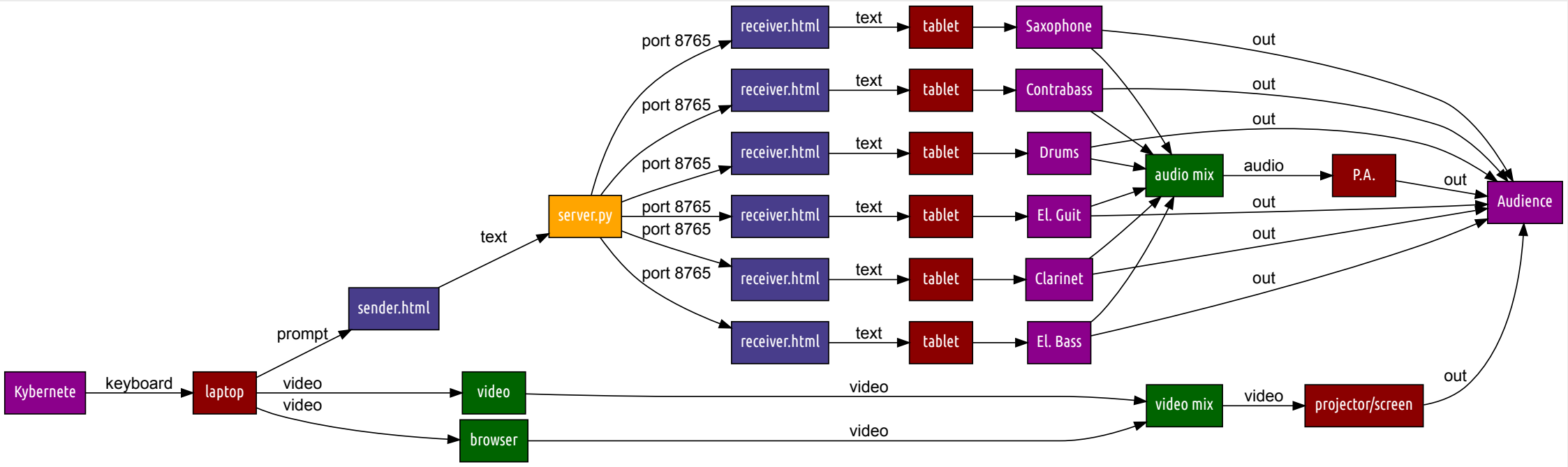


fig. Signal flow for *Nickelodeon digitale* — text, video and audio paths · code: github.com/vanderaalle/armillaria

RELAY → ALL RECEIVERS, UNFILTERED

```
{"text": "Saxophone: pianissimo, long", "from":
"Kybernete"}
```

same envelope whether "text" is broadcast or addressed to one name — the server doesn't parse it

ACK → ORIGINATING SENDER

```
{"type": "ack", "text": "..."}
```

+ "empty": true if no receivers are currently connected

SERVER

Envelope & acknowledgement

A Python `websockets` server relays every sender's message to *every* receiver, wrapped with the sender's identity. It never reads or filters the text — not even to tell a broadcast from a message addressed to one musician. Every relay also triggers an ack back to the kybernete, for continuous awareness of network state in the unreliable WiFi of live performance.

Selection happens downstream, entirely on the receiver's side — next.

RECEIVER

Selective addressing

`receiver.html?name=Saxophone`





sender / receiver The kybernete's input field (left) — a selective instruction reaching the Contrabass player (right)

03

NICKELODEON DIGITALE

NICKELODEON DIGITALE

— Early-cinema nickelodeons: storefront theatres, live musicians improvising accompaniment to silent film

— Altman's "living" performance space — music as variable, locally specific, never fixed in advance

— *Digitale* — digital, but also "of the finger": typing as the project's central performative gesture

BILDERATLAS SPRING BREAK

— A Warburgian *Bilderatlas* — an atlas of images — against the incongruity of *spring break*

— Three acts, ~20 min each — **Ieri**, **Oggi**, **Domani** — a non-linear history of humanity

— Found footage, Internet Archive & Wikimedia Commons — *original audio entirely removed*

ARTO FANTASMA

— A collective active since 2018, specialising in live sonorization of audiovisual material

— Clarinet (Edmondo Tedesco), saxophone (Simone Garino), drums (Dario Bruna), electric guitar (Carlo Barbagallo), electric bass (Andrea Valle), double bass (Federico Marchesano) — a groove-and-noise language that Armillaria destabilises and redirects live



fig. *Bilderatlas Spring Break* — Arto Fantasma, RKH Studio, Turin, 7 May 2026 · video: youtube.com/watch?v=ZhikoQiAcqA

IM-PROMPT-U

The kybernete does not program the music; the kybernete orients a generative device made up of natural agents — musicians who hear, feel, decide, misremember, and productively ignore.

13:05-14:04 - Zizek in conferenza

Time	Dest.	SEND
13:05	**ALL**	Philosophy conference after extinction - F minor - rubato - nervous thought, comic despair
13:18	Sax	Sax: thinking aloud, stuttering, brilliant and disgusting
13:32	Drums	Drums: nervous intellectual punctuation, coughs and table taps
13:45	Bass	Bass: anxious theoretical loop, unresolved
13:56	Clarinet	Clarinet: absurd little idea, almost a joke, then collapse

cue sheet Excerpt from *Žižek in conferenza* (*Domani*) — timecode, addressee, instruction

A TYPOLOGY OF INSTRUCTIONS

— **Scene-setters** — opening harmonic/affective orientation

— **Instrument roles** — selective, one musician at a time

— **Dramatic redirects** & **juxtaposition prompts**

"Keep smiling while the music becomes slightly inhuman" — *"cute pop over body horror"*

— **Negative constraints**, **transitions**, **stop/cut**

"Do not climax" — *"prepare for violent media intrusion"* — *"Stop. Leave a short, uncomfortable silence."*

04

CONCLUSIONS

WHAT REMAINS

> *The gap between what is typed and what is played is not noise to be minimised — it is the compositional substance of the work.*

OPEN DIRECTIONS

— **Stroma** — peer-to-peer variant, every node sends and receives, no fixed kybernete

— An ethnography of the interpretive gap — how performers read, interpret, productively misread

THANK YOU

b.surace@unito.it
andrea.valle@unito.it

CIM 2026 · Torino

Funded by the European Union — Next Generation EU, Italian Ministry of University and Research (MUR)
PNRR project IRACF (INTAFAM000055). Thanks to RKH Studio, Turin.